

# Write\_up\_TransformerLM

Problem (transformer\_accounting):

(a):The model has 1,640,452,800 trainable parameters, requiring about 6.56 GB of memory in float32 just for the parameters.

(b):3.52e12 FLOPs.

(c):FeedForwardNetwork

(d):As the model gets deeper and wider, the FFN and attention projection layers take up a larger share of the total FLOPs, since both scale roughly as  $\text{num\_layers } d_{\text{model}}^2$ . *The QK/attention-value products scale only as  $\text{num\_layers context\_length} * d_{\text{model}}$* , and the final vocabulary projection does not scale with the number of layers, so both become proportionally smaller in larger models.

(e):Increasing the context length from 1,024 to 16,384 increases the total forward-pass cost from about 3.52e12 to 1.34e14 FLOPs. The attention score and attention-value matrix multiplies become dominant because they scale quadratically with context length, while the projection, FFN, and LM-head costs scale only linearly with context length.